

OPC UA Pub/Sub App

OPC UA Pub/Sub for ctrlX OS

Version 4.6.0

Copyright

© Bosch Rexroth AG 2026-03

All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

Disclaimer

The specified data only serves to describe the product. As our products are constantly being further developed, no statements concerning a certain condition or suitability for a certain application can be derived from our information. The information given does not release the user from the obligation of own judgment and verification. Please note that our products are subject to a natural process of wear and aging.

DC-AE/PAG-SW (MiNi/PiaSt)

Table of contents

1	About this documentation	4
2	Important information	4
2.1	Product information	4
3	Version 4.6.0	4
3.1	New functions	4
3.1.1	Feature 1120337: OPC UA Pub/Sub – New functionalities . . .	4
3.1.2	Feature 1128807: OPC UA Pub/Sub – Extensions in the web interface	4
3.2	Resolved defects and modifications	4
3.2.1	Bug 1129483: OPC UA Pub/Sub – Error in the Pub/Sub configuration after restarting the control or switching the scheduler state	4
3.2.2	Bug 1155388: OPC UA Pub/Sub – Not all nodes were displayed in the Data Layer address space.	5
3.2.3	Bug 1169254: OPC UA Pub/Sub – Exception when loading a configuration, if "opcua_appconfig.cfg" in the app data is incorrect	5
3.2.4	Bug 1174875: Inconsistent behavior if the OPC UA PubSub Reader	5
4	Version 4.4.2	5
4.1	New functions	5
4.1.1	Feature 884438: OPC UA Pub/Sub – Supporting the Mirror Mode for DataSets of the subscriber	5
4.1.2	Feature 1003998: OPC UA Pub/Sub – Support for core24 and ctrlX OS version 4.X	5
4.1.3	Feature 1004007: OPC UA Pub/Sub – Upgrade of the OPC UA stack	5
4.1.4	Feature 1017133: OPC UA Pub/Sub – Improving the mechanism for loading the configuration	6
4.1.5	Feature 1054454: OPC UA Pub/Sub – App function extensions	6
4.1.6	Feature 1058015: OPC UA Pub/Sub – Extensions in the web interface	6
4.2	Resolved defects and modifications	6
4.2.1	Bug 1074551: OPC UA Pub/Sub – Writing a configuration via UaExpert failed	6
4.2.2	Bug 1084965: OPC UA Pub/Sub – The data types of the fields contained in the DataSets were only checked at runtime	6
4.2.3	Bug 1104802: OPC UA Apps – Display of the "OPC UA" navigation menu in the web interface despite missing authorization .	7
5	Service and support	8
6	Index	9

1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025 - 11	OPC UA Pub/Sub App Version 4.4.2
02	2026 - 3	OPC UA Pub/Sub App Version 4.6.0

Further information

- See ➔ [OPC UA Pub/Sub App, Application Manual](#)

2 Important information

2.1 Product information

For the latest information on our products, go to ➔ <https://www.boschrexroth.com>.

For current security advisories, please refer to ➔ <https://psirt.bosch.com/security-advisories/>

3 Version 4.6.0

3.1 New functions

3.1.1 Feature 1120337: OPC UA Pub/Sub – New functionalities

Description

The OPC UA PubSub provides the following new functionality:

- Status and configuration of the memory in the web interface.

3.1.2 Feature 1128807: OPC UA Pub/Sub – Extensions in the web interface

Description

The following new page is available in the web interface of the OPC UA Pub/Sub app:

- Reload and reset function for the configuration.

3.2 Resolved defects and modifications

3.2.1 Bug 1129483: OPC UA Pub/Sub – Error in the Pub/Sub configuration after restarting the control or switching the scheduler state

Description

Sporadically, an error was displayed in the pub/sub configuration after restarting the control or switching the scheduler state.

3.2.2 Bug 1155388: OPC UA Pub/Sub – Not all nodes were displayed in the Data Layer address space.

Description

The Data Layer nodes under the Data Layer address "opcuapubsub/address-space" were not displayed.

3.2.3 Bug 1169254: OPC UA Pub/Sub – Exception when loading a configuration, if "opcua_appconfig.cfg" in the app data is incorrect

Description

A fatal exception occurred in the Automation Core if "opcua_appconfig.cfg" was incorrect in the app data under opcuapubsub and a PubSub configuration was loaded via DL address "opcuapubsub/admin/load".

3.2.4 Bug 1174875: Inconsistent behavior if the OPC UA PubSub Reader

Description

The OPC UA PubSub-Publisher (Writer) and Subscriber (Reader) showed inconsistent behavior in the **SETUP** scheduler state. When switching from „OPERATING“ to „SETUP“, the publisher or subscriber continued to send or receive and the send or receive counter continued to increase accordingly. When switching from „SERVICE“ to „SETUP“, no data was sent or received.

Troubleshooting

When switching to „SETUP“, the publisher or subscriber starts sending or receiving, regardless of the previous scheduler state.

4 Version 4.4.2

4.1 New functions

4.1.1 Feature 884438: OPC UA Pub/Sub – Supporting the Mirror Mode for DataSets of the subscriber

Description

The OPC UA Pub/Sub app now supports the creation of subscribed DataSets in "MirrorMode".

With mirror-configured DataSets, the variables are automatically created as Data Layer RT variables in a separate MemoryMap. These RT variables can be read directly by other real-time applications on the ctrlX CORE control.

4.1.2 Feature 1003998: OPC UA Pub/Sub – Support for core24 and ctrlX OS version 4.X

Description

The OPC UA Pub/Sub app supports core24 and ctrlX OS version 4.X.

4.1.3 Feature 1004007: OPC UA Pub/Sub – Upgrade of the OPC UA stack

Description

The OPC UA Pub/Sub app uses the latest version of the OPC UA stack.

4.1.4 Feature 1017133: OPC UA Pub/Sub – Improving the mechanism for loading the configuration

Description

The loading of a configuration in OPC UA PubSub has been improved. Loading a configuration now works via the load program under Data Layer address "opcuaPubSub/configuration/load". This means that all configuration load processes can be tracked as Data Layer load tasks. In addition, the start and end time of the configuration load process is now specified in the Data Layer task.

4.1.5 Feature 1054454: OPC UA Pub/Sub – App function extensions

Description

New functions:

- Data from a Data Layer RT-MemoryMap can be published and subscribed to as a byte string via the "data" node.
- Improved logic for recognizing Data Layer RT nodes.

4.1.6 Feature 1058015: OPC UA Pub/Sub – Extensions in the web interface

Description

The following new pages are available in the web interface of the OPC UA Pub/Sub app:

- **Configuration:** Displaying the configuration processes carried out during the app runtime
- **Subscriber:** Current subscriber configuration.

4.2 Resolved defects and modifications

4.2.1 Bug 1074551: OPC UA Pub/Sub – Writing a configuration via UaExpert failed

Description

Under certain circumstances, writing a configuration via UaExpert failed with the error code "BadEncodingError". The error could only be rectified by restarting the OPC UA Server or ctrlX OS.

The cause of the error has been fixed.

4.2.2 Bug 1084965: OPC UA Pub/Sub – The data types of the fields contained in the DataSets were only checked at runtime

Description

In the OPC UA Pub/Sub, the data types of the individual DataSet fields were only checked at runtime. As a result, the fields continued to be written in every cycle until a type error was detected. After the error was detected, however, all fields of the DataSet were no longer transferred.

Bugfixing

The individual types are already checked during start-up and when loading a configuration, and the loading of the configuration is terminated with a corresponding error. In this case, no publishers or subscribers are active.

4.2.3 Bug 1104802: OPC UA Apps – Display of the "OPC UA" navigation menu in the web interface despite missing authorization

Description

In the OPC UA apps, the "OPC UA" navigation menu was displayed in the ctrlX web interface even though the current user session did not have the required OPC UA authorizations.

The cause of the error has been fixed.

5 Service and support

Our worldwide service network provides an optimized and efficient support. Our experts provide you with advice and assistance. You can contact us **24/7**.

Service Germany

Our technology-oriented Competence Center in Lohr, Germany, is responsible for all your service-related queries for electric drive and controls.

Contact the **Service Hotline** and **Service Helpdesk** under:

Phone: **+49 9352 40 5060**

Fax: **+49 9352 18 4941**

Email: ➔ service.svc@boschrexroth.de

Internet: ➔ <http://www.boschrexroth.com>

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

6 Index

H

Helpdesk 8

Hotline 8

I

Important information 4

P

Product information 4

S

Service hotline 8

Support 8

Bosch Rexroth AG
Bgm.-Dr.-Nebel-Str. 2
97816 Lohr a.Main
Germany
Tel. +49 9352 18 0
Fax +49 9352 18 8400
www.boschrexroth.com/electrics

